

Kazusa Tsubota

AWS AI Infrastructure / Platform Engineer

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Summary

Senior AI Infrastructure / AI Platform Engineer with 6+ years of backend and infrastructure experience, specializing in transforming traditional business systems into AI-powered production platforms and solving challenges across cost, performance, scalability, and operational complexity.

Experienced in designing scalable AI infrastructure with Kubernetes/EKS, GPU-based model serving, autoscaling, workload-based model routing, Terraform, Docker, and CI/CD. Strong hands-on experience across the AI production stack, including LLM inference, embeddings, Semantic Search, PostgreSQL/pgvector, RAG pipelines, and Tool Calling.

Combines infrastructure and backend engineering expertise to build reliable, scalable, observable, and cost-efficient AI platforms, from model serving and retrieval infrastructure to production APIs and deployment pipelines.

Experience

ScalyX.ai

AI Infrastructure / AI Production Engineer

Oct 2023 - Jun 2026 | USA | Remote

- Designed and operated infrastructure for a multi-tenant AI-powered retail platform, evolving the architecture from EC2-based workloads to Kubernetes on AWS EKS to support growing production workloads and future tenant expansion.
- Designed and optimized AI inference infrastructure by evaluating model characteristics, GPU requirements, and workload patterns, using appropriate GPU configurations and model-serving strategies to balance latency, scalability, availability, and infrastructure cost.
- Implemented workload-based multi-model routing, directing simple requests to lightweight models and complex requests to larger models, reducing unnecessary GPU consumption and contributing to a 73% reduction in monthly AWS infrastructure costs while maintaining service performance.
- Built production AI services using Python/FastAPI, PostgreSQL, and Docker, integrating LLM capabilities into backend services and business workflows.
- Implemented Semantic Search using Embeddings and PostgreSQL/pgvector, providing vector-based retrieval capabilities for AI applications.
- Designed and implemented RAG pipelines, connecting document/data ingestion, embedding generation, vector retrieval, and LLM generation to improve the accuracy and contextual relevance of AI responses.
- Implemented LLM Tool Calling to enable AI agents to interact with backend services and execute business operations through controlled application tools.
- Designed and operated scalable Kubernetes workloads using replicas, ALB-based load balancing, health checks, HPA, Node Autoscaler, and Helm to handle changing application and AI workloads.
- Implemented infrastructure as code using Terraform and optimized CI/CD pipelines with Docker, ECR, and Kubernetes to make AI application and infrastructure deployments more consistent, automated, and reliable.
- Built end-to-end observability using Prometheus, Grafana, OpenTelemetry, Jaeger, and CloudWatch, monitoring infrastructure, application, and AI workloads and reducing root-cause analysis time by 98%.
- Applied security controls across AWS and Kubernetes using IAM, VPC/private networking, Security Groups, Kubernetes RBAC, Secrets Manager, and tenant-level access controls.
- Performed load and performance testing to identify bottlenecks and validate system scalability, supporting workloads of up to 10K RPS.

Madoromi, Inc

Infrastructure / Backend Engineer

Apr 2020 - Jul 2023 | Tokyo, Japan | Remote

- Designed and operated cost-efficient AWS infrastructure for an early-stage product using EC2 and Aurora RDS, selecting

infrastructure capacity based on actual workload requirements rather than premature scaling.

- Designed and developed backend services using TypeScript/NestJS, PostgreSQL, and Node.js, building APIs and business logic for production applications.
- Designed a NestJS monorepo with Turborepo that consolidated multiple backend services onto a single EC2 instance, reducing infrastructure complexity and operational cost while maintaining sufficient capacity for the MVP workload.
- Implemented infrastructure automation using AWS Lambda and EventBridge to automatically start and stop development infrastructure according to engineering working hours, reducing unnecessary compute consumption.
- Optimized application and server resource usage to operate reliably within constrained EC2 CPU and memory resources.
- Introduced application and infrastructure observability to identify real production bottlenecks, using operational metrics to guide capacity planning and future architecture improvements.
- Worked across both backend development and infrastructure engineering, establishing a strong foundation in distributed application architecture, cloud infrastructure, deployment automation, and production operations.

Education

Kobe University

Bachelor of Engineering in Computer and Intelligent Systems Engineering | 2018

Skills & Abilities

AI Infrastructure & Platform

LLM Inference, Model Serving, GPU Infrastructure, Multi-Model Routing, Kubernetes, EKS, HPA, Node Autoscaling, Helm

AI Engineering

RAG, Embeddings, Semantic Search, Vector Search, PostgreSQL/pgvector, LLM Tool Calling, AI Agent Architecture

Cloud & Infrastructure

AWS, EC2, EKS, ALB, VPC, Private Subnets, IAM, Security Groups, RDS/Aurora, S3, ECR, CloudWatch

Infrastructure as Code & DevOps

Terraform, Docker, CI/CD, Infrastructure as Code, Infrastructure Automation, Helm, Deployment Optimization

Observability & Reliability

Prometheus, Grafana, OpenTelemetry, Jaeger, Metrics, Logs, Distributed Tracing, Performance Monitoring, Root Cause Analysis, Load Testing

Backend Engineering

Python, FastAPI, TypeScript, NestJS, Node.js, PostgreSQL, REST APIs, Monorepo, Turborepo

Security & Multi-Tenancy

IAM, Kubernetes RBAC, Network Isolation, Secrets Management, Tenant Isolation, Access Control

Performance & Cost Optimization

GPU Optimization, Resource Right-Sizing, Autoscaling, Workload-Based Routing, Capacity Planning, Performance Optimization, Cloud Cost Optimization